

Introduction to Trigonometry

Trigonometric ratios of acute angles, values for 0° , 30° , 45° , 60° and 90° , ratios of complementary angles and the three standard identities, with worked examples and practice questions.

Introduction

The word trigonometry comes from the Greek words trigon (triangle) and metron (measure). As the name suggests, it is the study of relationships between the angles and sides of a triangle. In Class 10, we deal only with right-angled triangles and acute angles (angles between 0° and 90°). The key idea is that for a given acute angle, the ratios of the sides of a right-angled triangle containing that angle are always the same, no matter how big or small the triangle is. Those ratios are called trigonometric ratios.

Trigonometric Ratios

Consider a right-angled triangle ABC, right-angled at B. For the acute angle at A, we name the sides like this:

- **Hypotenuse**, the side opposite the right angle (AC). It is always the longest side.
- **Opposite side** (or perpendicular), the side opposite to angle A (BC).
- **Adjacent side** (or base), the side that forms angle A together with the hypotenuse (AB).

If we look at the other acute angle C instead, then AB becomes the opposite side and BC becomes the adjacent side. The opposite and adjacent are always with respect to the angle you are talking about.

For the acute angle A, six ratios are defined:

Trigonometric Ratios for angle A

$$\sin A = \text{opposite} / \text{hypotenuse} = BC / AC$$

$$\cos A = \text{adjacent} / \text{hypotenuse} = AB / AC$$

$$\tan A = \text{opposite} / \text{adjacent} = BC / AB$$

$$\operatorname{cosec} A = \text{hypotenuse} / \text{opposite} = AC / BC = 1 / \sin A$$

$$\sec A = \text{hypotenuse} / \text{adjacent} = AC / AB = 1 / \cos A$$

$$\cot A = \text{adjacent} / \text{opposite} = AB / BC = 1 / \tan A$$

For $0^\circ < A < 90^\circ$, all six values are positive. The ratios depend only on the angle A, not on the size of the triangle.

Three facts are used very often, so it helps to remember them directly:

$$\tan A = \sin A / \cos A \quad \text{and} \quad \cot A = \cos A / \sin A \quad (\cos A \neq 0, \sin A \neq 0)$$

Also, $\sin A$ and $\cos A$ are always between 0 and 1 for an acute angle A (they can be 0 or 1 only at 0° or 90° , as we will see below), because in a right-angled triangle the hypotenuse is the longest side, so opposite/hypotenuse and adjacent/hypotenuse are always less than 1. But $\sec A$ and $\operatorname{cosec} A$ are always ≥ 1 , and $\tan A$ and $\cot A$ can be any positive number.

Worked Example

In $\triangle ABC$, right-angled at B, $AB = 3$ cm and $BC = 4$ cm. Find $\sin A$, $\cos A$ and $\tan A$.

First find the hypotenuse AC by Pythagoras: $AC^2 = AB^2 + BC^2 = 3^2 + 4^2 = 9 + 16 = 25$, so $AC = 5$ cm.

For angle A, opposite = $BC = 4$, adjacent = $AB = 3$, hypotenuse = 5.

$$\sin A = 4/5, \quad \cos A = 3/5, \quad \tan A = 4/3$$

You can check that $\tan A = \sin A / \cos A = (4/5) \div (3/5) = 4/3$.

Worked Example

If $\sin A = 1/2$, find $\cos A$ and $\tan A$, where A is acute.

Draw a right-angled triangle with opposite = 1 and hypotenuse = 2. Let adjacent = x . Then by Pythagoras, $x^2 + 1^2 = 2^2$, so $x^2 = 3$ and $x = \sqrt{3}$.

$$\cos A = \sqrt{3} / 2, \quad \tan A = 1 / \sqrt{3}$$

Trigonometric Ratios of Some Specific Angles

For most angles we need a calculator, but for 0° , 30° , 45° , 60° and 90° the ratios can be found exactly with simple geometry. NCERT expects you to know this table by heart.

How the values are found

45° comes from an isosceles right-angled triangle (both legs = 1, hypotenuse = $\sqrt{2}$).

30° and 60° come from an equilateral triangle of side 2, cut in half (sides 1, $\sqrt{3}$, 2).

0° and 90° are the limiting cases as the angle shrinks to 0° or grows to 90° in a right-angled triangle.

$\angle A$	0°	30°	45°	60°	90°
sin A	0	$1/2$	$1/\sqrt{2}$	$\sqrt{3}/2$	1
cos A	1	$\sqrt{3}/2$	$1/\sqrt{2}$	$1/2$	0
tan A	0	$1/\sqrt{3}$	1	$\sqrt{3}$	not defined (∞)
cosec A	not defined (∞)	2	$\sqrt{2}$	$2/\sqrt{3}$	1
sec A	1	$2/\sqrt{3}$	$\sqrt{2}$	2	not defined (∞)
cot A	not defined (∞)	$\sqrt{3}$	1	$1/\sqrt{3}$	0

Notice that $\sin A$ increases from 0 to 1 as A goes from 0° to 90° , while $\cos A$ decreases from 1 to 0. $\tan A$ increases from 0 and becomes not defined at 90° because $\cos 90^\circ = 0$ and $\tan A = \sin A / \cos A$ would mean division by zero.

Worked Example

Evaluate: $2 \sin^2 60^\circ + 3 \cos^2 45^\circ - \sec^2 30^\circ$.

Substitute from the table: $\sin 60^\circ = \sqrt{3}/2$, $\cos 45^\circ = 1/\sqrt{2}$, $\sec 30^\circ = 2/\sqrt{3}$.

$$2(\sqrt{3}/2)^2 + 3(1/\sqrt{2})^2 - (2/\sqrt{3})^2 = 2 \cdot (3/4) + 3 \cdot (1/2) - (4/3) = 3/2 + 3/2 - 4/3 = 3 - 4/3 = 5/3$$

Trigonometric Ratios of Complementary Angles

Two angles are complementary when they add to 90° . In a right-angled triangle, the two acute angles are always complementary. This gives a direct link between the ratios of an angle and its complement.

Complementary Angle Relations

For any acute angle A ($0^\circ < A < 90^\circ$):

$$\sin(90^\circ - A) = \cos A, \quad \cos(90^\circ - A) = \sin A$$

$$\tan(90^\circ - A) = \cot A, \quad \cot(90^\circ - A) = \tan A$$

$$\sec(90^\circ - A) = \operatorname{cosec} A, \quad \operatorname{cosec}(90^\circ - A) = \sec A$$

In words: the sine of an angle equals the cosine of its complement, tangent and cotangent swap, and secant and cosecant swap.

Worked Example

Evaluate $\sin 48^\circ - \cos 42^\circ$ without a calculator.

Since $42^\circ = 90^\circ - 48^\circ$, we have $\cos 42^\circ = \cos(90^\circ - 48^\circ) = \sin 48^\circ$. So $\sin 48^\circ - \cos 42^\circ = \sin 48^\circ - \sin 48^\circ = 0$.

Worked Example

Evaluate $\tan 26^\circ / \cot 64^\circ$.

$\cot 64^\circ = \cot(90^\circ - 26^\circ) = \tan 26^\circ$. So $\tan 26^\circ / \cot 64^\circ = \tan 26^\circ / \tan 26^\circ = 1$.

Trigonometric Identities

An **identity** is an equation that is true for all values of the angle for which both sides are defined. There are three fundamental identities that NCERT expects you to know and to be able to prove. All other 90° -type problems in the exercises reduce to these three with basic algebra.

Trigonometric Identities

For an acute angle A ($0^\circ \leq A \leq 90^\circ$) where the expressions are defined:

$$(1) \sin^2 A + \cos^2 A = 1$$

$$(2) 1 + \tan^2 A = \sec^2 A \quad (\text{for } 0^\circ \leq A < 90^\circ)$$

$$(3) 1 + \cot^2 A = \operatorname{cosec}^2 A \quad (\text{for } 0^\circ < A \leq 90^\circ)$$

Equivalently, (2) can be written as $\sec^2 A - \tan^2 A = 1$, and (3) as $\operatorname{cosec}^2 A - \cot^2 A = 1$.

Why (1) is true: In right $\triangle ABC$ with $\angle B = 90^\circ$ and $\angle A = A$, let $AB = x$, $BC = y$, $AC = r$. Then $\sin A = y/r$, $\cos A = x/r$, and by Pythagoras $x^2 + y^2 = r^2$. So $\sin^2 A + \cos^2 A = y^2/r^2 + x^2/r^2 = (x^2 + y^2)/r^2 = 1$. Identities (2) and (3) follow by dividing (1) by $\cos^2 A$ and $\sin^2 A$ respectively.

A very useful rearrangement of (1) is:

$$\sin^2 A = 1 - \cos^2 A, \quad \cos^2 A = 1 - \sin^2 A$$

These let you find one of $\sin A$ or $\cos A$ if you know the other.

Worked Example

If $\sin A = 3/5$, find $\cos A$ and verify $\sin^2 A + \cos^2 A = 1$.

Using the identity: $\cos^2 A = 1 - \sin^2 A = 1 - (3/5)^2 = 1 - 9/25 = 16/25$. Since A is acute, $\cos A$ is positive, so $\cos A = 4/5$. Check: $(3/5)^2 + (4/5)^2 = 9/25 + 16/25 = 25/25 = 1$.

Worked Example (Proving an identity)

Prove that $(1 + \tan^2 A) / (1 + \cot^2 A) = \tan^2 A$.

Left-hand side = $\sec^2 A / \operatorname{cosec}^2 A$ [using $1 + \tan^2 = \sec^2$ and $1 + \cot^2 = \operatorname{cosec}^2$]

$= (1/\cos^2 A) / (1/\sin^2 A) = \sin^2 A / \cos^2 A = \tan^2 A = \text{RHS}$

Tip for board exams: start from one side (usually the more complicated one) and use the identities to reach the other side. Avoid cross-multiplying unless you are sure neither side is zero.

Some identities are easier if you write everything in terms of $\sin A$ and $\cos A$ first. For example, to simplify $(\sec A + \tan A)(1 - \sin A)$, write $\sec A = 1/\cos A$ and $\tan A = \sin A / \cos A$, then combine.

Summary

- **Right-angled triangle:** for acute angle A , $\sin A = \text{opposite/hypotenuse}$, $\cos A = \text{adjacent/hypotenuse}$, $\tan A = \text{opposite/adjacent}$; $\operatorname{cosec} A = 1/\sin A$, $\sec A = 1/\cos A$, $\cot A = 1/\tan A$.
- **Relations:** $\tan A = \sin A / \cos A$, $\cot A = \cos A / \sin A$; ratios depend only on the angle, not the size of the triangle.
- **Specific values** for 0° , 30° , 45° , 60° , 90° as in the table above, to be memorised. In particular: $\sin 30^\circ = 1/2$, $\sin 45^\circ = 1/\sqrt{2}$, $\sin 60^\circ = \sqrt{3}/2$; $\cos 30^\circ = \sqrt{3}/2$, $\cos 45^\circ = 1/\sqrt{2}$, $\cos 60^\circ = 1/2$; $\tan 30^\circ = 1/\sqrt{3}$, $\tan 45^\circ = 1$, $\tan 60^\circ = \sqrt{3}$.
- **Complementary angles:** $\sin(90^\circ - A) = \cos A$, $\cos(90^\circ - A) = \sin A$, $\tan(90^\circ - A) = \cot A$, $\cot(90^\circ - A) = \tan A$, $\sec(90^\circ - A) = \operatorname{cosec} A$, $\operatorname{cosec}(90^\circ - A) = \sec A$.
- **Identities:** $\sin^2 A + \cos^2 A = 1$; $1 + \tan^2 A = \sec^2 A$; $1 + \cot^2 A = \operatorname{cosec}^2 A$.
- For acute A , $0 \leq \sin A \leq 1$ and $0 \leq \cos A \leq 1$, while $\sec A$, $\operatorname{cosec} A \geq 1$.

Practice Questions

1. In ΔPQR , right-angled at Q, $PQ = 5$ cm and $PR = 13$ cm. Find $\sin P$, $\cos P$ and $\tan P$. Also find $\sin R$ and $\cos R$ and verify that $\sin P = \cos R$.

2. If $\cos A = 12/13$, find $\sin A$ and $\tan A$, where A is an acute angle.

3. Evaluate without a calculator: $2 \tan^2 45^\circ + \cos^2 30^\circ - \sin^2 60^\circ$.

4. Evaluate: $\sin 25^\circ \cos 65^\circ + \cos 25^\circ \sin 65^\circ$.

5. Prove the identity: $(\operatorname{cosec} A - \sin A)(\sec A - \cos A) = 1 / (\tan A + \cot A)$.

6. Prove that $(\sin^4 A - \cos^4 A) / (\sin^2 A - \cos^2 A) = 1$.